

Exposé

Nusrat Jahan

I. MOTIVATION

The increasing reliance on neural networks in critical applications, particularly in healthcare, demands models that are not only accurate but also robust and well-calibrated. In tasks such as ECG and EEG classification, which directly influence medical diagnoses and patient outcomes, the consequences of incorrect or overconfident predictions can be severe. Misclassifying abnormal heart rhythms or brain signals with high confidence can lead to misdiagnoses, delayed treatments, or inappropriate interventions, ultimately putting patients at risk. This underscores the necessity of developing models that not only perform well but are also reliable under uncertain conditions.

While neural networks perform very well in controlled environments, their deployment in real-world scenarios introduces unpredictable challenges [1]. Time-series data, such as ECG and EEG signals, are often noisy and subject to distribution shifts [2] due to varying patient demographics, equipment conditions, or clinical settings [3]. These complexities require models that can handle variations and provide confidence estimates that accurately reflect how likely the predictions are to be correct.

A key motivation for this work is to address the gap in how current neural networks handle uncertainty and variability in time-series data. Although models can achieve high accuracy during training, they often fail when faced with inputs that deviate slightly from the training data [2], resulting in overconfident yet incorrect predictions. This lack of robustness and poor calibration can lead to life-threatening errors in healthcare settings, where reliability is critical.

Improving both robustness and calibration simultaneously is a challenging task. This work aims to explore and evaluate state-of-the-art techniques that tackle these issues for time-series data, specifically ECG and EEG classification. Techniques like Manifold Mixup can encourage neural networks to be less confident on unseen data by learning smoother decision boundaries [4], while ensemble methods and MIMO configurations can enhance reliability by aggregating multiple predictions [5] [6]. Despite the promise of these methods, their combined

effects on time-series data, particularly in ECG/EEG classification, remain underexplored. The central question is: Can a combination of these approaches result in a more reliable and well-calibrated model for time-series data?

The motivation for this work is to enhance robustness and calibration in deep learning models, making them more reliable in dynamic, real-world environments like healthcare. By exploring and combining techniques such as manifold Mixup, ensemble learning, and independent subnetworks, this work seeks to find the optimal solution for ECG/EEG classification.

II. PROBLEM FORMULATION

A. *Robustness in Neural Networks for Time-Series Data*

A major challenge in deploying neural networks for ECG/EEG classification is their susceptibility to noisy and unpredictable real-world inputs. In clinical practice, once a model is deployed, the distribution of observed data often shifts, deviating significantly from the original training distribution [2]. This phenomenon, known as distributional shift, introduces out-of-distribution (OOD) inputs [7], which can greatly affect the model's performance. For the safe and reliable deployment of machine learning systems in healthcare, ensuring robustness under these conditions is critical.

B. *Enhancing Calibration of Neural Networks:*

In addition to robustness, calibration (the ability of a model to provide reliable confidence estimates alongside predictions [8]) is equally important, particularly in high-stakes applications like healthcare. When faced with data shifts, traditional models frequently produce high-confidence predictions that do not accurately reflect the true likelihood of being correct, leading to overconfidence in noisy or OOD data [9]. This misalignment between model confidence and actual performance can have serious consequences in medical settings, where decisions depend heavily on the trustworthiness of the model's outputs.

To address these challenges, this work will explore several state-of-the-art methods aimed at improving both robustness and calibration in neural networks. One such

method, Manifold Mixup, trains models on interpolated hidden representations to smooth decision boundaries [4]. This approach has shown promise in enhancing a model's resilience to noise and adversarial examples. Additionally, ensemble learning methods will be considered for their ability to combine predictions from multiple models, which helps reduce the risk of critical misclassifications by aggregating diverse perspectives [10]. Finally, MIMO (Multi-Input Multi-Output) configurations allow for multiple independent subnetworks to be trained within a single model, offering the benefits of ensemble-like robustness without significantly increasing computational costs [6].

By combining these approaches, it may be possible to improve not only the robustness of neural networks but also their calibration, ensuring that confidence estimates are more closely aligned with the actual outcomes.

C. Research Questions:

This work will address the following key research questions:

- (1) How can we improve the robustness of neural networks to handle real-world input variations and noisy time-series data?
- (2) How can we improve the calibration of neural networks so that their confidence estimates better reflect the accuracy of their predictions, especially under challenging, real-world conditions?
- (3) What is the optimal combination of methods, such as Manifold Mixup, Ensemble learning, and MIMO configurations, that can provide the best trade-off between robustness and calibration for ECG/EEG classification?

III. TASKS AND PLANNING

A. Literature Research

The first task will be a comprehensive review of relevant literature, including:

- Methods for improving robustness in neural networks, focusing on adversarial attacks, input perturbations, and distribution shifts.
- Calibration techniques, including temperature scaling, ensemble methods, and Bayesian approaches.
- Evaluation of approaches like Manifold Mixup [4], Ensemble learning [11], and MIMO configurations [6], along with their prior successes and limitations in time-series applications.

B. Dataset and Preprocessing

This work will use publicly available ECG and EEG datasets.

- Preprocessing techniques will include normalisation, noise reduction, and handling missing or irregular data, ensuring that the CNN or transformer models can learn effectively from the time-series data.
- Evaluating the impact of modality-specific augmentations on the performance of CNN or transformer-based models for time-series data.

C. Model Development

The core task will be to design, and implement network models, focusing on combining different state-of-the-art methods:

- Implementation of CNN [12] or transformer [13] architectures for ECG/EEG classification, exploring their effectiveness for time-series data representation and feature extraction.
- Integrating Manifold Mixup into these architectures since mixup reduces the memorization of corrupt labels, increases the robustness [14].
- Applying ensemble learning techniques to aggregate predictions from multiple models, enhancing both robustness and calibration [10].
- Implementing MIMO configurations to leverage the power of independent subnetworks while reducing computational costs [6].

D. Model Evaluation

Evaluate model performance using metrics like accuracy, negative log-likelihood, and expected calibration error (ECE), along with robustness under distribution shifts and adversarial conditions.

E. Optimization and Tuning

Once the models are developed, the focus will be on optimizing their performance:

- Hyperparameter tuning for CNN and transformer models, including learning rate adjustments, dropout rates, and regularization strategies, to find optimal settings for robustness and calibration.
- Exploring trade-offs between model complexity, robustness, and calibration performance to find the best combination of methods for ECG/EEG classification.
- Performing cross-validation to ensure the models generalize well to unseen data.

F. Experiments and Analysis

Experiments will be conducted to evaluate each method (Ensemble, manifold mixup, etc.) individually and in combination to measure their impact on robustness and calibration.

G. Timeline

- Literature Research from 01.11.2024 until 30.11.2024
- Model Implementation and Preprocessing from 01.12.2024 until 31.12.2024
- Experiments with Ensemble Methods and Manifold Mixup from 01.01.2025 until 31.01.2025
- Combining Methods and Model Optimization from 01.02.2025 until 29.02.2025
- Analysis of Results and Model Refinement from 01.03.2025 until 31.03.2025
- Writing phase from 01.04.2025 until 30.04.2025

REFERENCES

- [1] Gunjan Chhabra, Sanjay Kumar, Avinash Kumar, Shrinwantu Raha, and Gonesh Saha. Analysis of deep learning in real-world applications: Challenges and progress. *Tuijin Jishu/Journal of Propulsion Technology*, 44:281–289, 07 2023.
- [2] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift, 2019.
- [3] Ziyu Liu, Azadeh Alavi, Minyi Li, and Xiang Zhang. Self-supervised contrastive learning for medical time series: A systematic review, 2023.
- [4] Vikas Verma, Alex Lamb, Christopher Beckham, Amir Najafi, Ioannis Mitliagkas, David Lopez-Paz, and Yoshua Bengio. Manifold mixup: Better representations by interpolating hidden states, 09–15 Jun 2019.
- [5] Thomas G. Dietterich. Ensemble methods in machine learning, 2000.
- [6] Marton Havasi, Rodolphe Jenatton, Stanislav Fort, Jeremiah Zhe Liu, Jasper Snoek, Balaji Lakshminarayanan, Andrew Mingbo Dai, and Dustin Tran. Training independent subnetworks for robust prediction, 2021.
- [7] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in ai safety, 2016.
- [8] Kaiyue Wu, Changwu Huang, and Xin Yao. Confidence estimation based on the explanations of model’s predictions, 2023.
- [9] Angela Zhang, Lei Xing, James Zou, and Joseph C. Wu. Shifting machine learning for healthcare from development to deployment and from models to data, Dec 2022.
- [10] Tanjim Mahmud, Anik Barua, Dilshad Islam, Mohammad Shahadat Hossain, Rishita Chakma, Koushick Barua, Mahabuba Monju, and Karl Andersson. Ensemble deep learning approach for ecg-based cardiac disease detection: Signal and image analysis, 2023.
- [11] Trung Trinh, Markus Heinonen, Luigi Acerbi, and Samuel Kaski. Input-gradient space particle inference for neural network ensembles, 2024.
- [12] Taissir Fekih Romdhane, Haikel Alhichri, Ridha Ouni, and Mohamed Atri. Electrocardiogram heartbeat classification based on a deep convolutional neural network and focal loss. *Computers in Biology and Medicine*, 123:103866, 2020.
- [13] Akhil Vaid, Joy Jiang, Ashwin Sawant, Stamatios Lerakis, Edgar Argulian, Yuri Ahuja, Joshua Lampert, Alexander Charney, Hayit Greenspan, Jagat Narula, Benjamin Glicksberg, and Girish N. Nadkarni. A foundational vision transformer improves diagnostic performance for electrocardiograms, Jun 2023.
- [14] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization, 2018.